

Tab 1

Question

1. Do we need to run every example in both models to compare result

H

2. Agent history is from where?
3. Market info is 2 all the time?
4. Do the examples need to be spread out within the levels? The test examples we choose are too extreme they are either 1 and 5

We can do 2, 3, 4 level too

1st prompt

Agent_Instruction_Prompt = f"""

You are performing a dataset-relative social-media popularity LEVEL prediction task.

Your goal is to predict the target post's popularity_level using the provided image, metadata, and labeled examples.

This is not an image-quality rating task.

This is not a general social-media virality task.

This is a dataset-relative classification task calibrated by the labeled examples.

Prediction target

The required output is popularity_level, an integer from 1 to 5.

The original continuous label is popularity_score, a log-view score ranging approximately from 1.00 to 16.56.

The popularity_level is derived from popularity_score using this exact mapping:

Level 1: popularity_score in [1.00, 4.50), bottom 20%

Level 2: popularity_score in [4.50, 5.70), 20th-40th percentile

Level 3: popularity_score in [5.70, 6.90), 40th-60th percentile

Level 4: popularity_score in [6.90, 8.60), 60th-80th percentile

Level 5: popularity_score in [8.60, 16.56], top 20%

Important:

Level 1 is a broad low-popularity bin. Many visually clear, technically acceptable, or ordinary photos can still be Level 1.

Do not assign Level 2 or above just because the image is clear, colorful, well-focused, or has descriptive tags.

Calibration principle

Use the labeled examples as the main source of calibration.

Internally compare the target post to the most similar labeled examples, especially examples with the same or similar:

category

subcategory

concept

subject matter

visual style

title and tag pattern

user profile metadata

temporal and spatial metadata

The predicted level should follow the levels of the closest labeled examples, not general assumptions about social media.

General social-media intuition must be weaker than the dataset examples.

Conservative bin rule

When the evidence is mixed, choose the lower level unless there is strong evidence from similar labeled examples for a higher level.

For ordinary niche subjects, such as common wildlife, fish, plants, food items, everyday objects, or routine travel scenes, do not move above Level 1 unless the post clearly resembles labeled examples in Level 2 or higher.

A clear subject and accurate tags are positive evidence, but they are usually not enough to cross from Level 1 to Level 2.

Evidence to consider

Evaluate the post using:

Image content

main subject and visual clarity

composition, color, lighting, and technical quality

emotional, narrative, humorous, unusual, or novelty appeal

broad public appeal versus niche audience appeal

whether the post resembles ordinary low-level examples or distinctive high-level examples

Text metadata

whether the title is complete or cut off

whether the title is specific, generic, or promotional

whether tags are useful, repetitive, niche, or generic

do not assume many tags automatically increase popularity

Category context

category, subcategory, and concept

compare primarily with similar examples from the same category or subcategory when available

do not compare a niche post against unrelated high-performing posts unless no closer examples exist

Temporal and spatial metadata

post datetime

location metadata quality

treat blank latitude, blank longitude, 0 latitude, 0 longitude, or geoaccuracy 0 as missing or unreliable unless the metadata clearly provides a valid location

do not treat missing or unreliable location as a real geographic coordinate

User profile metadata

photo count

Pro status

available user description and account history

do not infer followers, reputation, account reach, or previous engagement

Additional metadata

public status and media status only indicate availability

do not treat public/ready status as strong popularity evidence

Rules against unsupported assumptions

Do not assume that:

better image quality means higher popularity_level

specific tags automatically improve engagement

location words automatically improve engagement

high photo count means high reach

Pro membership means high reach

missing metadata automatically lowers popularity

a niche but nice-looking photo should be Level 2 or Level 3

a professional-looking photo must be above Level 1

Do not invent:

views

likes

comments

shares

followers

recommendation exposure

account reputation

external popularity

Output requirements

Return valid JSON only.

Use exactly this structure:

```
{
  "popularity_level": 1,
  "reason": "The post has a clear subject and some descriptive metadata, which are positive signals. However, compared with the closest labeled examples, it remains an ordinary or niche dataset item with limited broad emotional, narrative, or novelty appeal. Since the evidence is not strong enough to cross into a higher bin, Level 1 is the most appropriate prediction."
}
```

Requirements:

popularity_level must be an integer from 1 to 5.

reason must contain 2 to 4 concise sentences.

The reason must mention both positive and negative evidence.

The reason must distinguish visual quality from dataset-relative popularity.
Do not output popularity_score.
Do not output estimated_score_range.
Do not return markdown, bullet points, or any text outside the JSON object.

Labeled examples

{examples}

Target Post metadata:

Image reference

"image_path": "train/61884@N47/749030.jpg",
"row_index": 273638

Category

"Category": "Entertainment",
"Concept": "amc",
"Pid": "749030",
"Uid": "61884@N47",
"Subcategory": "Movies"

Text

"Alltags": "ny newyork sexy celebrity rabbit art hat fashion animal brooklyn asian coneyisland japanese star tv google king artist dragon god manhattan famous gothic goth uma ufo pop taxidermy vogue cnn tuxedo bikini tophat unitednations playboy entertainer oddities genius mermaid amc mardigras salvadordali performer unicorn billclinton seamonster billgates aol vangogh curiosities sideshow jeffkoons pennstation globalwarming mart magician takashimurakami pablopicasso steampunk damienhirst cryptozoology freakshow seara immortalized takeshiyamada roguetaxidermy searabbit barrackobama ladygaga climategate ????? manwithrabbit",

"Pid": "749030",
"Uid": "61884@N47",
"Mediatype": "photo",
"Title": "Dr. Takeshi Yamada and Seara (Coney Island Sea Rabbit at the New York Penn Station in Manhattan, NY on May 13, 2015. 20150513 145=C4BW"

Temporal and spatial metadata

"Postdate": "1430536538",
"Uid": "61884@N47",
"Pid": "749030",
"Longitude": "",
"Geoaccuracy": "0",
"Latitude": "0",
"Postdate_datetime": "2015-05-01 23:15:38"

User profile

"Uid": "61884@N47",
"Pid": 749030,
"photo_firstdate": null,
"photo_count": 5712.0,
"ispro": 0.0,

"timezone_offset": "-05:00",

"photo_firstdatetaken": "2003-01-01 00:01:00",

"timezone_id": "Eastern Time (US & Canada)",

"user_description": "takeshiyamada.weebly.com/ The Sea Rabbit (*Monafluffchus americanus*) of Coney Island, Brooklyn, New York – This unique sea-dwelling rabbit, which is actually a close relative of the sea lion, was officially discovered and investigated by Henry Hudson when he first visited this land to colonize the area by order of the Dutch government. It was named New Amsterdam -- today's New York City. This island was named after he saw the beach covered with strange swimming wild rabbits. The word "Coney Island" means "wild rabbit island" in Dutch (originally Conyne Eylandt, or Konijneneiland in modern Dutch spelling). Sea rabbits were also referred mermaid rabbit, merrabbit, rabbit fish or seal rabbit in the natural history documents in the 17th century. The current conservation status, or risk of extinction, of the sea rabbit is Extinct in the Wild. This website features two species of sea rabbits, which have been taken care of by Dr. Takeshi Yamada (????) at the Coney Island Sea Rabbit Repopulation Center, which is a part of the Marine biology department of the Coney Island University in Brooklyn, New York. They are – Coney Island Sea Rabbit (*Monafluffchus americanus*) called "Seara" and Coney Island Tiger-striped Sea Rabbit (*Monafluffchus konjinicus*) called "Stripes". The photographs and videos featured in this website chronicle adventures of the Coney Island sea rabbits and the world as seen by them. This article also documented efforts of Dr. Takeshi Yamada for bringing back the nearly extinct sea rabbits to Coney Island in the City of New York and beyond. Dr. Yamada produced a series of public lectures, workshops, original public live interactive fine art performances and fine art exhibitions about sea rabbits at a variety of occasions and institutions in the City of New York and beyond. Dr. Yamada is an internationally active educator, book author, wildlife conservationist and high profile artist, who lives and works in Brooklyn, New York.

----- Sea Rabbit Other
Common Names: Coney Island Sea Rabbit, Beach Rabbit, Seal Rabbit, mer-rabbit, merrabbit, Atlantic Sea Rabbit. Latin Name: *Monafluffchus americanus* Origin: Atlantic coast of the United States Description of the specimen: In the early 17th century's European fur craze drove the fleet of Dutch ships to the eastern costal area of America. Then Holland was the center of the world just like the Italy was in the previous century. New York City was once called New Amsterdam when Dutch merchants landed and established colonies. Among them, Henry Hudson is probably the most recognized individual in the history of New York City today. "This small island is inhabited by two major creatures which we do not have in our homeland. The one creature is a large arthropod made of three body segments: the frontal segment resembles a horseshoe, the middle segment resembles a spiny crab and its tail resembles a sharp sword. Although they gather beaches here in great numbers, they are not edible due to their extremely offensive odor. Another creature which is abundant here, has the head of wild rabbit. This animal of great swimming ability has frontal legs resemble the webbed feet of a duck. The bottom half of the body resembles that of a seal. This docile rabbit of the sea is easy to catch as it does not fear people. The larger male sea rabbits control harems of 20 to 25 females. The meat of the sea rabbit is very tender and tasty." This is what Hadson wrote in his personal journal in 1609 about the horseshoe crab and the sea rabbit in today's Coney Island area of Brooklyn, New York. Sadly, just like the Dodo bird and the Thylacine, the sea rabbit was driven to extinction by the European settlers' greed. When Dutch merchants and traders arrived here, sea rabbits were one of the first animals they hunted down to bring their furs to homeland to satisfy the fur craze of the time. To increase

the shipment volume of furs of sea rabbit and beavers from New Amsterdam, Dutch merchants also started using wampum (beads made of special clam shells) as the first official currency of this country. At the North Eastern shores of the United States, two species of sea rabbits were commonly found. They are Coney Island Sea Rabbit (*Monafluffchus americanus*) and Coney Island Tiger-striped Sea Rabbit (*Monafluffchus konjinicus*). Sadly, due to their over harvesting in the previous centuries, their conservation status became "Extinct in the Wild" (ET) in the Red List Endangered species by the International Union for Conservation of Nature (IUCN). Currently, these sea rabbits are only found at breeding centers at selected zoos and universities such as Coney Island Aquarium and Coney Island University in Brooklyn, New York. The one shown in this photograph was named "Seara" and has been cared by Dr. Takeshi Yamada at Coney Island University. The sea rabbit is one of the families of the Pinniped order. Pinnipeds (from Latin penna = flat and pes/pedis = foot) are sea-mammals: they are homeothermic (i.e having high and regulated inner temperature), lung-breathing (i.e dependant on atmospheric oxygen) animals having come back to semi aquatic life. As soon as they arrive ashore, females are caught by the nearest adult male. Males can maintain harems of about 20 females on average. Several hours to several days after arriving ashore, pregnant females give birth to eight to ten pups with a dark brown fur. As soon as birth occurs, the mother's special smell and calls help her pups bond specifically to her. The mother stays ashore with her pup for about one week during which the pup gains weight. During the first week spent with her newborn, the mother becomes receptive. She will be impregnated by the bull, which control the harem. Implantation of the embryo will occur 3 months later, in March-April. During the reproductive period, the best males copulate with several tens females. To do so, males have to stay ashore without feeding in order to keep their territory and their harem. In mid-January, when the last females have been fecundated, males leave at sea to feed. Some of them will come back later in March-April for the moult. The other ones will stay at sea and will come back on Coney Island only in next November. After fecundation, the mother goes at sea for her first meal. At sea, mothers feed on clams, crabs, shrimps, fish (herring, anchovy, Pollock, capelin etc.) and squids. When she is back, the mother recovers her pups at the beach she left them. Suckling occurs after auditive and olfactory recognition had occured. In March-April, the dark brown fur is totally replaced by an adult-like light brownish grey fur during the moult that lasts 1-2 months. This new fur is composed by 2 layers. Externally, the guard fur is composed by flat hairs that recover themselves when wet. By doing so, they make a water-proof barrier for the under fur. The underfur retains air when the seal is dry. Because of isolating properties of the air, the underfur is the insulating system of the fur. In March-April, the fur of adults is partially replaced. First reproduction occurs at 1-yr old in females. Males are physiologically matures at 1 year old but socially matures at +2 years old. NOTE: The name of Coney Island is commonly thought to be derived from the Dutch Konijn Eylandt or Rabbit Island as apparently the 17th century European settlers noted many rabbits running amuck on the island. <http://www.takeshiyamada.weebly.com/performances.html> <http://www.takeshiyamada.weebly.com/performances.html> <http://www.takeshiyamada.weebly.com/sea-rabbit-center.html> <http://www.takeshiyamada.weebly.com/sea-rabbit-center.html> ===== <https://www.flickr.com/photos/searabbit22> <https://www.flickr.com/photos/searabbit22> <https://www.flickr.com/photos/searabbit21> <https://www.flickr.com/photos/searabbit21> <https://www.flickr.com/photos/searabbit20> <https://www.flickr.com/photos/searabbit20>

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<https://www.flickr.com/photos/takeshiyamadapaintings/>
>www.flickr.com/photos/takeshiyamadapaintings/

===== For any questions, please email
contact Takeshi Yamada, Art & Rogue Taxidermy, Museum of World Wonders, official
website. <a href= <http://www.takeshiyamada.weebly.com/> rel= nofollow
>www.takeshiyamada.weebly.com/

===== <a href=
<http://www.takeshiyamada.weebly.com> rel= nofollow >www.takeshiyamada.weebly.com
For any questions, please contact Dr. Takeshi Yamada. His email address is posted in the
chapter page (the last page or the first page). (Updated December 22, 2014)",
"location_description": null

Additional metadata

"Mediastatus": "ready",
"Pathalias": "searabbits23",
"Ispublic": "1",
"Pid": "749030",
"Uid": "61884@N47"

2nd prompt

Agent_Instruction_Prompt = f"""

You are performing a dataset-relative social-media popularity-level prediction task.

Your goal is to predict the target post's `popularity\_level` using the provided image, metadata, and labeled training examples.

This is not an image-quality rating task.

This is not a general social-media virality task.

This is a supervised classification task calibrated by the training examples.

Prediction target

Predict `popularity\_level` as an integer from 1 to 5.

The original continuous label is `popularity\_score`, a log-view score ranging approximately from 1.00 to 16.56.

Use this exact mapping:

- Level 1: popularity_score in [1.00, 4.50)
- Level 2: popularity_score in [4.50, 5.70)
- Level 3: popularity_score in [5.70, 6.90)
- Level 4: popularity_score in [6.90, 8.60)
- Level 5: popularity_score in [8.60, 16.56]

Do not predict the continuous score. Predict only the corresponding integer level.

Step 1: Calibrate carefully from the training examples

Before evaluating the target post, study the labeled training examples carefully.

Use the examples to infer how this dataset connects popularity level with:

- category and subcategory
- subject matter
- visual content
- title and tags
- user profile
- location information
- temporal metadata
- missing metadata

Do not rely mainly on general assumptions about social media.

Identify internally the training examples that are most similar to the target, especially examples with similar:

- category
- subcategory
- concept
- subject
- visual style
- user profile
- location metadata
- title and tag pattern

The labels of the closest training examples should be the main source of calibration.

When the training examples conflict with general intuition, follow the training examples.

Do not assume that a clear, colorful, attractive, professional, or well-tagged post must have a high popularity level.

Step 2: Evaluate the target using three rubrics

After calibrating from the training examples, evaluate the target using the following three rubrics.

Rubric A: Content

Evaluate:

- main subject
- category, subcategory, and concept
- visual clarity
- composition
- lighting and color
- technical image quality
- novelty and distinctiveness
- emotional, humorous, narrative, aspirational, or unusual appeal
- broad public appeal versus niche audience appeal
- whether the image looks ordinary, repetitive, staged, commercial, documentary, or personal
- title completeness and relevance
- title specificity
- tag relevance
- tag repetition
- promotional or commercial language
- similarity to the content of labeled training examples

Important rules:

- Image quality alone must not determine the level.

- Many ordinary but clear images can still be Level 1.
- Many tags do not automatically increase popularity.
- Descriptive tags do not automatically improve engagement.
- Promotional content is not automatically positive or negative.
- Use the training examples to determine how content features affect the level.

Rubric B: User profile

Evaluate:

- photo count
- Pro status
- user description
- location description
- available account history
- first-photo dates
- path alias
- whether the account appears personal, commercial, professional, organizational, or brand-based
- similarity to user profiles in the labeled training examples

Important rules:

- Do not infer follower count.
- Do not infer account reach.
- Do not infer reputation.
- Do not infer previous engagement.
- Do not assume that a high photo count means high popularity.
- Do not assume that Pro status means high popularity.
- Do not assume that a non-Pro user must have low popularity.
- Treat user-profile features as evidence only when similar patterns appear in the training examples.

Rubric C: Location and temporal metadata

Evaluate:

- latitude and longitude
- geoaccuracy
- location description
- timezone
- consistency between location fields
- whether the location is meaningful for the category or subject
- post date and time
- whether similar location and temporal patterns appear in the training examples

Important rules:

- Blank coordinates, zero coordinates, or geoaccuracy 0 should be treated as missing or unreliable.
- Precise coordinates do not automatically increase popularity.
- Missing location does not automatically reduce popularity.
- Use location and time only as supporting evidence unless the training examples show a strong relationship.

Step 3: Combine the evidence

Combine the three rubrics instead of letting one feature dominate.

Internally follow this process:

1. Find the most similar labeled training examples.
2. Compare the target's Content rubric with those examples.
3. Compare the target's User Profile rubric with those examples.
4. Compare the target's Location and Temporal rubric with those examples.
5. Determine which popularity level best matches the complete evidence pattern.
6. Check that the selected level is consistent with the exact score-bin mapping.

The closest labeled examples should outweigh unsupported assumptions.

Do not default automatically to Level 1.

Do not default automatically to Level 3.

Do not choose a level merely because it feels safe.

Choose the level best supported by the training examples and the combined rubric.

Rules against unsupported assumptions

Do not invent:

- views
- likes
- comments
- shares
- followers
- account reach
- recommendation exposure
- account reputation
- external popularity

Do not assume that:

- better image quality always means higher popularity
- niche content must be Level 1
- attractive people or fashion content must be highly popular
- specific tags automatically increase engagement
- precise location automatically increases engagement

- public or ready status strongly affects popularity

Labeled training examples

{examples}

Target post

```
## Target Post metadata:
### Image reference
  "image_path": "train/69099@N68/822183.jpg",
  "row_index": 280894
### Category
  "Category": "Entertainment",
  "Concept": "pop",
  "Pid": "822183",
  "Uid": "69099@N68",
  "Subcategory": "Music"
### Text
  "Alltags": "lighting new girls portrait people woman hot color colour
cute sexy celebrity art me mannequin girl beautiful beauty face fashion
female pose indonesia asian fun photography model eyes nikon highheel
highheels photographer dof dress bokeh modeling supermodel femme fine
creative dramatic naturallight exhibition pop modelo event human
jakarta attractive stunning celebrities serene wardrobe manual
fotografia fashionshow popular mode modelling glamorous finrart
fotomodel photomodel manualexposure strobist nikkor70200mm
wanitaindonesia nikond800 wanitacantik wanitakarir",
  "Pid": "822183",
  "Uid": "69099@N68",
  "Mediatype": "photo",
  "Title": "PHOTO HUNTING POPULAR MAGAZINE"

### Temporal and spatial metadata
  "Postdate": "1456522478",
  "Uid": "69099@N68",
  "Pid": "822183",
  "Longitude": "",
  "Geoaccuracy": "0",
  "Latitude": "0",
  "Postdate_datetime": "2016-02-26 16:34:38"

### User profile
  "Uid": "69099@N68",
```

```

    "Pid": 822183,
    "photo_firstdate": null,
    "photo_count": 2577.0,
    "ispro": 0.0,
    "timezone_offset": null,
    "photo_firstdatetaken": "2011-02-26 15:01:00",
    "timezone_id": null,
    "user_description": "I am a Professional Photographer living in
Jakarta, Indonesia. I am proficient in all kinds of Photography. My
professional experience include wedding photography, commercial and
corporate photography, photography in a multitude of events and
photography performed in studio and on location. I would like to thank
those of you who have spent time to visit my photo stream. I hope you
will enjoy my photography works. Warm regards GULFAN AFERO",
    "location_description": null

### Additional metadata

    "Mediastatus": "ready",
    "Pathalias": "gulfanaferophotography",
    "Ispublic": "1",
    "Pid": "822183",
    "Uid": "69099@N68"

```

Required output

Return valid JSON only.

Use exactly this structure:

```

{{
  "popularity_level": 1
}}
```

Requirements:

- `popularity\_level` must be an integer from 1 to 5.
- Do not output `popularity\_score`.
- Do not output an explanation.
- Do not output markdown.
- Do not include a trailing comma.
- Do not output any text outside the JSON object.

"""

3rd prompt

Agent_Instruction_Prompt = f"""

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- Level 3: popularity_score in [5.70, 6.90)
- Level 4: popularity_score in [6.90, 8.60)
- Level 5: popularity_score in [8.60, 16.56]

The levels represent dataset-relative popularity ranges.

Level 1 is a broad low-popularity bin. A visually clear, colorful, well-composed, or technically acceptable post can still belong to Level 1.

Do not assign Level 2 or higher only because the image is attractive, well-focused, professionally staged, or accompanied by descriptive tags.

Primary example-relative calibration rule

The predicted `popularity\_level` must be directly calibrated to the labeled training examples provided in this prompt.

The examples define the practical meaning of Levels 1 through 5 for this dataset.

Do not interpret the levels mainly through general social-media intuition.

Before selecting a level, carefully study the labeled examples and identify the examples that are most similar to the target in:

- category
- subcategory

- concept
- subject matter
- visual content and style
- title pattern
- tag pattern
- user-profile metadata
- location metadata
- temporal metadata
- missing-data pattern

The labels of the closest examples must be the primary source of calibration.

The final predicted level should be relevant to and consistent with the levels assigned to the closest examples.

Use the following rules:

- If the target closely resembles several examples from one level, prefer that level.
- If the target falls between examples from two adjacent levels, choose the level supported by the closest overall examples.
- Do not assign a different level from highly similar examples unless the target has meaningful differences in content, user profile, or location.
- If general intuition conflicts with the labels of similar examples, follow the examples.
- Do not allow unrelated high-performing examples to dominate the comparison when closer category-specific examples are available.

Required internal evaluation process

Before producing the answer, internally follow these steps:

1. Study the labeled training examples.
2. Identify the closest examples to the target.
3. Calibrate the likely level from those examples.
4. Evaluate the target using the Content rubric.
5. Evaluate the target using the User Profile rubric.
6. Evaluate the target using the Location and Temporal rubric.
7. Combine the three rubrics.
8. Compare the combined evidence with the closest examples again.
9. Select the level that is most consistent with the example-based calibration.
10. Check that the selected level follows the exact score-bin mapping.

Do not output this internal evaluation.

Rubric A: Content

Evaluate the target's content using the image, category, title, and tags.

Consider:

- main subject
- category
- subcategory
- concept
- visual clarity
- composition
- lighting
- color
- technical quality
- subject distinctiveness
- novelty
- emotional appeal
- narrative appeal
- humorous or unusual elements
- aspirational or lifestyle appeal
- broad public appeal versus niche audience appeal
- whether the image appears ordinary, repetitive, staged, personal, documentary, commercial, or promotional
- whether the title is complete or cut off
- whether the title is specific, generic, descriptive, or promotional
- whether the tags match the image
- whether the tags are broad, niche, repetitive, descriptive, or promotional
- similarity to the content patterns in the labeled examples

Important content rules:

- Image quality alone must not determine the level.
- A clear or professional-looking image can still be Level 1.
- An ordinary-looking image is not automatically Level 1.
- Many tags do not automatically increase popularity.
- Descriptive tags do not automatically improve engagement.
- Repetitive tags do not automatically improve discoverability.
- Promotional content is not automatically positive or negative.
- Attractive people, fashion, travel, sports, or lifestyle content is not automatically high-popularity content.
- Use the labeled examples to determine how content characteristics relate to popularity levels.

Rubric B: User profile

Evaluate the available user-profile information.

Consider:

- photo count

- Pro status
- user description
- location description
- first-photo date
- first-photo-taken date
- timezone
- path alias
- available account history
- whether the account appears personal, professional, commercial, organizational, or brand-based
- similarity to user-profile patterns in the labeled examples

Important user-profile rules:

- Do not infer follower count.
- Do not infer account reach.
- Do not infer account reputation.
- Do not infer recommendation exposure.
- Do not infer previous engagement.
- Do not assume that high photo count means high popularity.
- Do not assume that low photo count means low popularity.
- Do not assume that Pro status means high popularity.
- Do not assume that non-Pro status means low popularity.
- Do not assume that a commercial account is automatically popular.
- Treat user-profile characteristics as meaningful only when similar patterns appear in the labeled examples.

Rubric C: Location and temporal metadata

Evaluate the available location and temporal information.

Consider:

- latitude
- longitude
- geoaccuracy
- location description
- timezone
- timezone offset
- consistency among location fields
- whether the location is meaningful for the category or subject
- post date
- post time
- similarity to location and temporal patterns in the labeled examples

Important location and temporal rules:

- Blank latitude or longitude should be treated as missing.

- Latitude or longitude equal to 0 should be treated as missing or unreliable unless the metadata clearly indicates a valid coordinate.
- Geoaccuracy 0 should be treated as missing or unreliable.
- Precise coordinates do not automatically increase popularity.
- Location-related words in the tags do not automatically increase popularity.
- Missing location does not automatically reduce popularity.
- Posting at a particular time does not automatically increase or decrease popularity.
- Use location and time mainly as supporting evidence unless the labeled examples show a strong pattern.

Combining the rubrics

Combine Content, User Profile, and Location and Temporal evidence.

Do not let one isolated feature control the prediction.

Use this decision rule:

1. Begin with the levels of the closest labeled examples.
2. Compare the target's content with those examples.
3. Compare the target's user profile with those examples.
4. Compare the target's location and temporal metadata with those examples.
5. Determine whether the combined differences justify moving to an adjacent level.
6. Do not move to a higher or lower level without meaningful example-based evidence.
7. Select the level that best matches the complete evidence pattern.

When the three rubrics point toward different levels, give the greatest weight to the labeled examples that most closely match the target across all three rubrics.

Do not automatically choose the lowest level.

Do not automatically choose the middle level.

Do not choose a level merely because it appears safe.

Rules against unsupported assumptions

Do not assume that:

- better image quality always means higher popularity
- a clear subject means higher popularity
- a visually attractive image must be above Level 1
- niche content must be Level 1
- many tags automatically improve engagement
- specific tags automatically improve engagement
- location words automatically improve engagement

- precise location automatically improves engagement
- high photo count means strong reach
- Pro membership means strong reach
- missing metadata automatically lowers popularity
- public status strongly affects popularity
- ready media status strongly affects popularity
- commercial or promotional content must be popular
- a professional-looking post must be above Level 1

Do not invent:

- views
- likes
- comments
- shares
- followers
- account reach
- recommendation exposure
- account reputation
- external popularity
- prior engagement

Category-specific comparison rule

The target belongs to:

- Category: Travel&Active&Sports
- Subcategory: Skateboarding
- Concept: skater

Prioritize examples in this order when available:

1. Skateboarding examples
2. Sports and active-lifestyle examples
3. Beach, outdoor, travel, or fashion-lifestyle examples
4. Commercial or brand-oriented examples with similar content
5. Other examples with similar user-profile and location patterns

Do not calibrate the target mainly against unrelated categories when closer examples are available.

Labeled training examples

{examples}

Target post

Image reference

Image path: train/53515@N9/649591.jpg
Row index: 262636

Category

Category: Travel&Active&Sports
Subcategory: Skateboarding
Concept: skater

Text

Title: Black Sport Shorts

Tags:

summer, portrait, black, sexy, beach, portugal, nature, fashion, sport, canon, girlfriend, goddess, style, beachlife, bikini, thong, skate, femalemodel, skater, canon5d, shorts, brunette, mermaid, sensuality, beachgirl, siren, aveiro, hotgirl, daring, skatergirl, blackbikini, praiadabarra, sexygirl, bikinigirl, hotwife, portuguese girl, ef135f2, sexygirlfriend, beachstyle, thongbikini, brazilianbikini, sexybikini, minibikini, bikinifashion, sportshorts, ef135, girlsinsport, beachlook, bikinilover, canon5diii, beachsoul, sportivelook, daringgirl, mypiki, sportivebikini, sportivegirl

Tag pattern:

- many tags
- strongly repetitive
- promotional
- focused on bikini and swimwear branding
- includes beach and Portugal-related terms
- includes fashion and modeling terms
- includes skateboarding and sports terms
- includes Canon photography-equipment terms

Media type: photo

Temporal and spatial metadata

Post datetime: 2015-09-19 06:11:47
Latitude: 40.641080
Longitude: -8.75260
Geoaccuracy: 16
Timezone offset: +00:00
Timezone: GMT: Dublin, Edinburgh, Lisbon, London
Location description: Aveiro, Portugal

User profile

Photo count: 25
Pro member: no
First photo upload date: missing
First photo capture date: 2014-01-01 00:00:00
Path alias: mypiki

User description:

MYPIKI is a Portuguese bikini brand. The profile promotes bold and distinctive swimwear products and includes commercial branding, product descriptions, a website, and a contact email.

Additional metadata

Public: yes
Media status: ready

Final decision check

Before returning the answer, internally verify that:

1. The selected level is consistent with the closest labeled examples.
2. The prediction reflects the combined Content, User Profile, and Location and Temporal rubrics.
3. The prediction does not rely mainly on general social-media intuition.
4. The prediction does not rely on unsupported assumptions.
5. The selected level follows the exact popularity-score bin mapping.
6. A different level would require meaningful differences from the closest examples.

Output requirements

Return valid JSON only.

Use exactly this structure:

```
{{  
  "popularity_level": 1  
}}
```

Requirements:

- `popularity\_level` must be an integer from 1 to 5.
- Do not output `popularity\_score`.
- Do not output an estimated score range.
- Do not output an explanation.
- Do not output markdown.

- Do not include a trailing comma.
 - Do not output any text outside the JSON object.
- ""

4th prompt

Target post

Image reference

"image_path": "train/69099@N68/822183.jpg",
"row_index": 280894

Category

"Category": "Entertainment",
"Concept": "pop",
"Pid": "822183",
"Uid": "69099@N68",
"Subcategory": "Music"

Text

"Alltags": "lighting new girls portrait people woman hot color colour
cute sexy celebrity art me mannequin girl beautiful beauty face fashion
female pose indonesia asian fun photography model eyes nikon highheel
highheels photographer dof dress bokeh modeling supermodel femme fine
creative dramatic naturallight exhibition pop modelo event human
jakarta attractive stunning celebrities serene wardrobe manual
fotografia fashionshow popular mode modelling glamorous finrart
fotomodel photomodel manualexposure strobist nikkor70200mm
wanitaindonesia nikond800 wanitacantik wanitakarir",
"Pid": "822183",
"Uid": "69099@N68",
"Mediatype": "photo",
"Title": "PHOTO HUNTING POPULAR MAGAZINE"

Media type: photo

Temporal and spatial metadata

"Postdate": "1456522478",
"Uid": "69099@N68",
"Pid": "822183",
"Longitude": "",
"Geoaccuracy": "0",
"Latitude": "0",
"Postdate_datetime": "2016-02-26 16:34:38"

User profile

```

"UId": "69099@N68",
"PId": 822183,
"photo_firstdate": null,
"photo_count": 2577.0,
"ispro": 0.0,
"timezone_offset": null,
"photo_firstdatetaken": "2011-02-26 15:01:00",
"timezone_id": null,
"user_description": "I am a Professional Photographer living in
Jakarta, Indonesia. I am proficient in all kinds of Photography. My
professional experience include wedding photography, commercial and
corporate photography, photography in a multitude of events and
photography performed in studio and on location. I would like to thank
those of you who have spent time to visit my photo stream. I hope you
will enjoy my photography works. Warm regards GULFAN AFERO",
"location_description": null
### Additional metadata

"Mediastatus": "ready",
"Pathalias": "gulfanaferophotography",
"Ispublic": "1",
"PId": "822183",
"UId": "69099@N68"

```

Without reason:

```
Agent_Instruction_Prompt = f"""
```

```
You are performing a dataset-relative social-media popularity-level
prediction task.
```

```
Your goal is to predict the target post's `popularity_level` using:
```

1. the labeled training examples,
2. the target image and text,
3. the target user profile,
4. the target location and temporal metadata.

```
This is not an image-quality rating task.
```

```
This is not a general social-media virality task.
```

```
This is a supervised classification task whose labels must be
calibrated from the provided examples.
```

```
## Prediction target
```

Predict `popularity\_level` as one integer from 1 to 5.

The original continuous label is `popularity\_score`, a log-view score ranging approximately from 1.00 to 16.56.

Use this exact mapping:

- Level 1: popularity_score in [1.00, 4.50)
- Level 2: popularity_score in [4.50, 5.70)
- Level 3: popularity_score in [5.70, 6.90)
- Level 4: popularity_score in [6.90, 8.60)
- Level 5: popularity_score in [8.60, 16.56]

Predict only the integer level. Do not predict the continuous score.

Primary calibration rule

The labeled training examples define the practical meaning of Levels 1 through 5 for this dataset.

The target must be evaluated relative to those examples, not mainly through general assumptions about social-media popularity.

Before selecting a level:

1. Identify the labeled examples most similar to the target.
2. Compare the target with those examples using the three required rubrics below.
3. Use the labels of the closest examples as the starting point.
4. Move to a different level only when meaningful differences across the rubrics justify the change.

The final prediction must be consistent with the closest examples.

When general intuition conflicts with the labeled examples, follow the labeled examples.

Similarity priority

Identify the closest examples using the following evidence:

- category

- subcategory
- concept
- subject matter
- visual content and style
- title and tag pattern
- user-profile characteristics
- location and temporal metadata
- missing-data pattern

Prioritize examples in this order when available:

1. Same subcategory or concept
2. Same category with similar content
3. Similar user-profile and metadata pattern
4. Similar visual, textual, or commercial style
5. Broader examples only when closer matches are unavailable

Do not let unrelated high- or low-level examples dominate the comparison.

Required evaluation rubrics

Evaluate the target strictly through all three rubrics.

Rubric A: Content

Evaluate:

- category, subcategory, concept, and main subject
- visual clarity and technical quality
- composition, lighting, and color
- novelty and visual distinctiveness
- emotional, humorous, narrative, unusual, aspirational, or lifestyle appeal
- broad appeal versus niche appeal
- whether the content appears ordinary, repetitive, personal, staged, commercial, documentary, or promotional
- title completeness, specificity, and relevance
- tag relevance, repetition, breadth, and promotional character
- similarity to the content patterns of the closest labeled examples

Content rules:

- Image quality alone must not determine the level.
- A visually strong image can still have a low popularity level.
- An ordinary image is not automatically Level 1.
- Many tags, descriptive tags, or promotional wording do not automatically increase popularity.
- Judge content features by how they correspond to the labeled examples.

Rubric B: User Profile

Evaluate:

- photo count
- Pro status
- user description
- location description
- first-photo dates
- available account history
- timezone
- path alias
- whether the account appears personal, professional, commercial, organizational, or brand-based
- similarity to user profiles in the closest labeled examples

User-profile rules:

- Do not infer followers, reach, reputation, recommendation exposure, or previous engagement.
- Do not assume that photo count, Pro status, or commercial identity directly determines popularity.
- Use profile characteristics only when comparable patterns appear in the labeled examples.

Rubric C: Location and Temporal Metadata

Evaluate:

- latitude and longitude
- geoaccuracy
- location description
- timezone and timezone offset
- consistency among location fields
- relevance of location to the subject

- post date and time
- similarity to location and temporal patterns in the closest labeled examples

Location and temporal rules:

- Blank coordinates, zero coordinates, and geoaccuracy 0 should be treated as missing or unreliable unless clearly valid.
- Precise location does not automatically increase popularity.
- Missing location does not automatically reduce popularity.
- Posting time should be supporting evidence unless the examples show a clear relationship.

Decision procedure

Internally follow this process:

1. Find the closest labeled examples.
2. Establish an initial candidate level from their labels.
3. Evaluate the target under Content.
4. Evaluate the target under User Profile.
5. Evaluate the target under Location and Temporal Metadata.
6. Compare the complete rubric pattern with the closest examples.
7. Decide whether the differences justify staying at the candidate level or moving to an adjacent level.
8. Select the level best supported by the examples and the combined rubrics.
9. Verify that the final level follows the exact score-bin mapping.

When the rubrics conflict, give the greatest weight to the labeled examples that are most similar to the target across all three rubrics.

Do not automatically choose Level 1.

Do not automatically choose Level 3.

Do not choose a level merely because it appears conservative or safe.

Unsupported assumptions

Do not invent or estimate:

- views
- likes
- comments

- shares
- followers
- account reach
- account reputation
- recommendation exposure
- previous engagement
- external popularity

Do not treat public status or ready media status as strong popularity evidence.

Labeled training examples

{examples}

Target post

Target post

Image reference

"image_path": "train/50564@N82/1111339.jpg",
"row_index": 304995

Category

"Category": "Food",
"Concept": "dinner",
"Pid": "1111339",
"Uid": "50564@N82",
"Subcategory": "General"

Text

"Alltags": "newzealand food apple dinner roadtrip september iatethis
iphone newplymouth 2015 appleiphone iphone6 chestnutgrey sarahoettli
appleiphone6 september2015 roadtrip2015 19september2015",
"Pid": "1111339",
"Uid": "50564@N82",
"Mediatype": "photo",
"Title": "Dinner"

Temporal and spatial metadata

"Postdate": "1442660508",
"Uid": "50564@N82",

```
"Pid": "1111339",
"Longitude": "",
"Geoaccuracy": "0",
"Latitude": "0",
"Postdate_datetime": "2015-09-19 07:01:48"

### User profile
"UId": "50564@N82",
"PId": 1111339,
"photo_firstdate": null,
"photo_count": 7426.0,
"ispro": 1.0,
"timezone_offset": null,
"photo_firstdatetaken": "2003-11-09 10:17:01",
"timezone_id": null,
"user_description": null,
"location_description": null

### Additional metadata
"Mediastatus": "ready",
"Pathalias": "sarah_oettli",
"Ispublic": "1",
"PId": "1111339",
"UId": "50564@N82"

## Final consistency check

Before answering, verify internally that:

1. The prediction follows the closest labeled examples.
2. All three rubrics were considered.
3. No unsupported assumptions affected the decision.
4. Any movement away from the closest examples is justified by
meaningful rubric differences.
5. The selected level follows the exact score-bin mapping.

## Required output

Return valid JSON only.

Use exactly this structure:
```

```
{{  
  "popularity_level": 1  
}}
```

Requirements:

- `popularity\_level` must be an integer from 1 to 5.
 - Do not output `popularity\_score`.
 - Do not output an explanation.
 - Do not output markdown.
 - Do not include a trailing comma.
 - Do not output any text outside the JSON object.
- """

With reason:

```
Agent_Instruction_Prompt = f"""
```

You are performing a dataset-relative social-media popularity-level prediction task.

Your goal is to predict the target post's `popularity\_level` using:

1. the labeled training examples,
2. the target image and text,
3. the target user profile,
4. the target location and temporal metadata.

This is not an image-quality rating task.

This is not a general social-media virality task.

This is a supervised classification task whose labels must be calibrated from the provided examples.

Prediction target

Predict `popularity\_level` as one integer from 1 to 5.

The original continuous label is `popularity\_score`, a log-view score ranging approximately from 1.00 to 16.56.

Use this exact mapping:

- Level 1: popularity_score in [1.00, 4.50)
- Level 2: popularity_score in [4.50, 5.70)
- Level 3: popularity_score in [5.70, 6.90)
- Level 4: popularity_score in [6.90, 8.60)
- Level 5: popularity_score in [8.60, 16.56]

Predict only the integer level. Do not predict the continuous score.

Primary calibration rule

The labeled training examples define the practical meaning of Levels 1 through 5 for this dataset.

The target must be evaluated relative to those examples, not mainly through general assumptions about social-media popularity.

Before selecting a level:

1. Identify the labeled examples most similar to the target.
2. Compare the target with those examples using the three required rubrics below.
3. Use the labels of the closest examples as the starting point.
4. Move to a different level only when meaningful differences across the rubrics justify the change.

The final prediction must be consistent with the closest examples.

When general intuition conflicts with the labeled examples, follow the labeled examples.

Similarity priority

Identify the closest examples using the following evidence:

- category
- subcategory
- concept
- subject matter
- visual content and style
- title and tag pattern
- user-profile characteristics
- location and temporal metadata

- missing-data pattern

Prioritize examples in this order when available:

1. Same subcategory or concept
2. Same category with similar content
3. Similar user-profile and metadata pattern
4. Similar visual, textual, or commercial style
5. Broader examples only when closer matches are unavailable

Do not let unrelated high- or low-level examples dominate the comparison.

Required evaluation rubrics

Evaluate the target strictly through all three rubrics.

Rubric A: Content

Evaluate:

- category, subcategory, concept, and main subject
- visual clarity and technical quality
- composition, lighting, and color
- novelty and visual distinctiveness
- emotional, humorous, narrative, unusual, aspirational, or lifestyle appeal
- broad appeal versus niche appeal
- whether the content appears ordinary, repetitive, personal, staged, commercial, documentary, or promotional
- title completeness, specificity, and relevance
- tag relevance, repetition, breadth, and promotional character
- similarity to the content patterns of the closest labeled examples

Content rules:

- Image quality alone must not determine the level.
- A visually strong image can still have a low popularity level.
- An ordinary image is not automatically Level 1.
- Many tags, descriptive tags, or promotional wording do not automatically increase popularity.
- Judge content features by how they correspond to the labeled examples.

Rubric B: User Profile

Evaluate:

- photo count
- Pro status
- user description
- location description
- first-photo dates
- available account history
- timezone
- path alias
- whether the account appears personal, professional, commercial, organizational, or brand-based
- similarity to user profiles in the closest labeled examples

User-profile rules:

- Do not infer followers, reach, reputation, recommendation exposure, or previous engagement.
- Do not assume that photo count, Pro status, or commercial identity directly determines popularity.
- Use profile characteristics only when comparable patterns appear in the labeled examples.

Rubric C: Location and Temporal Metadata

Evaluate:

- latitude and longitude
- geoaccuracy
- location description
- timezone and timezone offset
- consistency among location fields
- relevance of location to the subject
- post date and time
- similarity to location and temporal patterns in the closest labeled examples

Location and temporal rules:

- Blank coordinates, zero coordinates, and geoaccuracy 0 should be treated as missing or unreliable unless clearly valid.
- Precise location does not automatically increase popularity.
- Missing location does not automatically reduce popularity.
- Posting time should be supporting evidence unless the examples show a clear relationship.

Decision procedure

Internally follow this process:

1. Find the closest labeled examples.
2. Establish an initial candidate level from their labels.
3. Evaluate the target under Content.
4. Evaluate the target under User Profile.
5. Evaluate the target under Location and Temporal Metadata.
6. Compare the complete rubric pattern with the closest examples.
7. Decide whether the differences justify staying at the candidate level or moving to an adjacent level.
8. Select the level best supported by the examples and the combined rubrics.
9. Verify that the final level follows the exact score-bin mapping.

When the rubrics conflict, give the greatest weight to the labeled examples that are most similar to the target across all three rubrics.

Do not automatically choose Level 1.

Do not automatically choose Level 3.

Do not choose a level merely because it appears conservative or safe.

Unsupported assumptions

Do not invent or estimate:

- views
- likes
- comments
- shares
- followers
- account reach
- account reputation
- recommendation exposure
- previous engagement

- external popularity

Do not treat public status or ready media status as strong popularity evidence.

Labeled training examples

{examples}

Target post

Target post

Image reference

"image_path": "train/55361@N36/495018.jpg",
"row_index": 243383

Category

"Category": "Family",
"Concept": "infant",
"Pid": "495018",
"Uid": "55361@N36",
"Subcategory": "Babies"

Text

"Alltags": "bear family woman baby female mom toy person kid infant
child sleep birth mother pregnancy pregnant belly maternity parent
newborn crib cradle childbirth",
"Pid": "495018",
"Uid": "55361@N36",
"Mediatype": "photo",
"Title": "Pregnant Woman"

Temporal and spatial metadata

"Postdate": "1455687769",
"Uid": "55361@N36",
"Pid": "495018",
"Longitude": "",
"Geoaccuracy": "0",

```
"Latitude": "0",
"Postdate_datetime": "2016-02-17 00:42:49"

### User profile
  "Uid": "55361@N36",
  "Pid": 495018,
  "photo_firstdate": null,
  "photo_count": 184.0,
  "ispro": 0.0,
  "timezone_offset": "+01:00",
  "photo_firstdatetaken": "2000-01-01 00:03:40",
  "timezone_id": "Sarajevo, Skopje, Warsaw, Zagreb",
  "user_description": "Free stock photos for both personal and
commercial use.",
  "location_description": "Olsztyn, Poland"

### Additional metadata
  "Mediastatus": "ready",
  "Pathaliases": "freestocks",
  "Ispublic": "1",
  "Pid": "495018",
  "Uid": "55361@N36"

## Final consistency check

Before answering, verify internally that:

1. The prediction follows the closest labeled examples.
2. All three rubrics were considered.
3. No unsupported assumptions affected the decision.
4. Any movement away from the closest examples is justified by
meaningful rubric differences.
5. The selected level follows the exact score-bin mapping.
6. The reasoning clearly explains why the selected level is more
appropriate than the nearby alternatives.
7. The reasoning refers to the closest labeled examples and the three
rubrics.

## Required output

Return valid JSON only.

Use exactly this structure:
```

```
{{
  "popularity_level": 1,
  "reason": "Provide 2 to 4 concise sentences explaining why this level
was selected. Refer to the closest labeled examples and summarize the
strongest evidence from Content, User Profile, and Location and
Temporal Metadata. Distinguish visual quality from dataset-relative
popularity and avoid unsupported assumptions."
}}
```

Requirements:

- `popularity\_level` must be an integer from 1 to 5.
 - Do not output `popularity\_score`.
 - Do not output an explanation.
 - Do not output markdown.
 - Do not include a trailing comma.
 - Do not output any text outside the JSON object.
 - `reason` must contain 2 to 4 concise sentences.
 - The reasoning must explain how the target compares with the closest labeled examples.
 - The reasoning must address the three rubrics: Content, User Profile, and Location and Temporal Metadata.
 - The reasoning should emphasize the most influential rubric evidence rather than listing every field.
 - The reasoning must explain why the selected level is appropriate relative to nearby levels.
 - Do not invent unavailable engagement or account information.
- """

5th prompt

Without reason

Agent_Instruction_Prompt = f"""

You are performing a dataset-relative social-media popularity-level prediction task.

Your goal is to predict the target post's `popularity\_level` using:

1. the labeled training examples,
2. the target image and text,
3. the target user profile,
4. the target location and temporal metadata.

This is not an image-quality rating task.

This is not a general social-media virality task.

This is a supervised classification task calibrated strictly from the provided labeled examples.

Prediction target

Predict `popularity\_level` as one integer from 1 to 5.

The original continuous label is `popularity\_score`, a log-view score ranging approximately from 1.00 to 16.56.

Use this exact mapping:

- Level 1: popularity_score in [1.00, 4.50)
- Level 2: popularity_score in [4.50, 5.70)
- Level 3: popularity_score in [5.70, 6.90)
- Level 4: popularity_score in [6.90, 8.60)
- Level 5: popularity_score in [8.60, 16.56]

Predict only the integer level.

Do not predict the continuous score.

Primary calibration rule

The labeled training examples define the practical meaning of Levels 1 through 5 for this dataset.

The target must be evaluated relative to those examples, not mainly through general assumptions about social-media popularity.

Follow this order strictly:

1. Identify the labeled examples most similar to the target in category, subcategory, concept, subject matter, and image presentation.
2. Check the actual popularity levels of those examples.
3. Determine the two most plausible adjacent levels for the target.
4. Compare the target with the closest examples from both candidate levels.
5. Evaluate the comparison using Content, User Profile, and Location and Temporal Metadata.
6. Select the level whose examples match the target more closely across the complete evidence pattern.

When general intuition conflicts with the labeled examples, follow the labeled examples.

Similarity priority

Identify the closest examples in this order:

1. Same or closely related concept and subcategory
2. Same category with similar subject matter and image presentation
3. Similar title and tag pattern
4. Similar user-profile characteristics
5. Similar location, temporal, and missing-data pattern
6. Broader examples only when closer examples are unavailable

Content similarity must be considered before user-profile similarity.

An example with similar Pro status, photo count, account age, or commercial identity is not necessarily close if its category, concept, subject, or presentation differs substantially from the target.

Do not let unrelated high- or low-level examples dominate the comparison.

Required evaluation rubrics

Evaluate the target through all three rubrics.

Rubric A: Content

Evaluate:

- category, subcategory, concept, and main subject
- visual clarity and technical quality
- composition, lighting, and color
- ordinary versus distinctive presentation
- novelty and visual distinctiveness
- emotional, humorous, narrative, unusual, aspirational, or lifestyle appeal
- broad appeal versus niche appeal
- whether the content appears repetitive, personal, staged, commercial, documentary, or promotional
- title completeness, specificity, and relevance
- tag relevance, repetition, breadth, and promotional character
- similarity to the content patterns of the closest labeled examples

Content rules:

- Content similarity is the primary basis for selecting the closest examples.
- Image quality alone must not determine the level.
- A visually strong image can still have a low or middle popularity level.
- An ordinary image is not automatically Level 1.
- Many tags, descriptive tags, or promotional wording do not automatically increase popularity.
- A positive visual feature should affect the predicted level only when it helps distinguish the plausible candidate levels in the labeled examples.
- Do not treat professional lighting, emotional subject matter, attractive composition, or broad appeal as automatic evidence for Level 4 or Level 5.

Rubric B: User Profile

Evaluate:

- photo count

- Pro status
- user description
- location description
- first-photo dates
- available account history
- timezone
- path alias
- whether the account appears personal, professional, commercial, organizational, or brand-based
- similarity to user profiles in the closest labeled examples

User-profile rules:

- User-profile evidence is secondary to content similarity.
- Do not infer followers, reach, reputation, recommendation exposure, or previous engagement.
- Do not describe a profile as strong, established, influential, high-performing, or authoritative solely because it has Pro status, a high photo count, or a long account history.
- User-profile evidence may affect the prediction only when:
 1. the compared examples are already similar in content,
 2. the profile differences correspond to different labels,
 3. the target clearly matches one side of that comparison.
- Otherwise, treat user-profile metadata as non-decisive.

Rubric C: Location and Temporal Metadata

Evaluate:

- latitude and longitude
- geoaccuracy
- location description
- timezone and timezone offset
- consistency among location fields
- relevance of location to the subject
- post date and time
- similarity to location and temporal patterns in the closest labeled examples

Location and temporal rules:

- Blank coordinates, zero coordinates, and geoaccuracy 0 should be treated as missing or unreliable unless clearly valid.
- Precise location does not automatically increase popularity.
- Missing location does not automatically reduce popularity.
- Posting time should be supporting evidence unless the labeled examples show a clear relationship.
- Location and temporal metadata should refine the comparison, not replace content-based calibration.

Contrastive ordinal decision procedure

Internally follow this process:

1. Find the closest labeled examples based primarily on category, concept, subject matter, and image presentation.
2. Record the actual popularity levels of those examples.
3. Determine the two most plausible adjacent levels.
4. Identify the closest examples from each candidate level.
5. Compare the target with both candidate levels using all three rubrics.
6. Separate descriptive features from label-separating features.
7. Treat a feature as predictive only when it distinguishes the candidate levels in the labeled examples.
8. Begin from the level of the closest overall examples.
9. Test movement only one level at a time.
10. Do not skip an intermediate level.
11. For each upward movement, verify that the target is more similar to the higher-level examples than to the lower-level examples.
12. Select the level that best matches the complete evidence pattern.

When the rubrics conflict:

- Give the greatest weight to content similarity and the actual levels of the closest examples.
- Use user-profile and location evidence only to refine the comparison.
- Do not raise the level mainly because the account is old, Pro, highly active, commercial, or associated with many uploaded photos.
- Do not choose the level with the most positive-sounding description.
- Choose the level whose examples are closest overall.

Level-boundary rule

The levels are ordinal dataset bins, not ratings of how good or attractive the post is.

Level 5 represents the highest dataset-relative bin, not simply a professional, emotional, attractive, searchable, or commercially useful post.

Before selecting Level 5:

1. Compare the target directly with the closest Level 4 and Level 5 examples.
2. Check whether the target is more similar to Level 5 across the complete evidence pattern.
3. Do not select Level 5 when the same positive characteristics also appear in closely matched Level 4 examples without a clear distinguishing difference.

Apply the same adjacent-level comparison when deciding between any two neighboring levels.

Non-causal interpretation rule

Do not explain or infer popularity through hypothetical mechanisms such as:

- searchability
- discoverability
- algorithmic exposure
- stock-photo demand
- free-use licensing
- brand reach
- account authority
- profile strength
- typical social-media engagement
- recommendation exposure

The task is example-relative classification, not causal prediction.

Unsupported assumptions

Do not invent or estimate:

- views

- likes
- comments
- shares
- followers
- account reach
- account reputation
- recommendation exposure
- previous engagement
- external popularity

Do not treat public status or ready media status as strong popularity evidence.

Labeled training examples

{examples}

Target post

{target_post}

Final consistency check

Before answering, verify internally that:

1. The closest examples were selected primarily by content and category similarity.
2. The actual levels of the closest examples were checked.
3. Two plausible adjacent levels were compared directly.
4. All three rubrics were considered.
5. Descriptive features were not automatically treated as popularity evidence.
6. Pro status, photo count, account age, commercial identity, or stock-photo identity did not independently raise the prediction.
7. No causal claims about discoverability, engagement, reach, or account authority affected the decision.
8. No intermediate level was skipped.
9. The selected level follows the exact score-bin mapping.

Required output

Return valid JSON only.

Use exactly this structure:

```
{{  
  "popularity_level": 1  
}}
```

Requirements:

- `popularity\_level` must be an integer from 1 to 5.
- Do not output `popularity\_score`.
- Do not output an explanation or reasoning.
- Do not output markdown.
- Do not include a trailing comma.
- Do not output any text outside the JSON object.

"""

With reason

Agent_Instruction_Prompt = f"""

You are performing a dataset-relative social-media popularity-level prediction task.

Your goal is to predict the target post's `popularity\_level` using:

1. the labeled training examples,
2. the target image and text,
3. the target user profile,
4. the target location and temporal metadata.

This is not an image-quality rating task.

This is not a general social-media virality task.

This is a supervised classification task calibrated strictly from the provided labeled examples.

Prediction target

Predict `popularity\_level` as one integer from 1 to 5.

The original continuous label is `popularity\_score`, a log-view score ranging approximately from 1.00 to 16.56.

Use this exact mapping:

- Level 1: popularity_score in [1.00, 4.50)
- Level 2: popularity_score in [4.50, 5.70)
- Level 3: popularity_score in [5.70, 6.90)
- Level 4: popularity_score in [6.90, 8.60)
- Level 5: popularity_score in [8.60, 16.56]

Predict only the integer level.

Do not predict the continuous score.

Primary calibration rule

The labeled training examples define the practical meaning of Levels 1 through 5 for this dataset.

The target must be evaluated relative to those examples, not mainly through general assumptions about social-media popularity.

Follow this order strictly:

1. Identify the labeled examples most similar to the target in category, subcategory, concept, subject matter, and image presentation.
2. Check the actual popularity levels of those examples.
3. Determine the two most plausible adjacent levels for the target.
4. Compare the target with the closest examples from both candidate levels.

5. Evaluate the comparison using Content, User Profile, and Location and Temporal Metadata.
6. Select the level whose examples match the target more closely across the complete evidence pattern.

When general intuition conflicts with the labeled examples, follow the labeled examples.

Similarity priority

Identify the closest examples in this order:

1. Same or closely related concept and subcategory
2. Same category with similar subject matter and image presentation
3. Similar title and tag pattern
4. Similar user-profile characteristics
5. Similar location, temporal, and missing-data pattern
6. Broader examples only when closer examples are unavailable

Content similarity must be considered before user-profile similarity.

An example with similar Pro status, photo count, account age, or commercial identity is not necessarily close if its category, concept, subject, or presentation differs substantially from the target.

Do not let unrelated high- or low-level examples dominate the comparison.

Required evaluation rubrics

Evaluate the target through all three rubrics.

Rubric A: Content

Evaluate:

- category, subcategory, concept, and main subject
- visual clarity and technical quality
- composition, lighting, and color
- ordinary versus distinctive presentation
- novelty and visual distinctiveness
- emotional, humorous, narrative, unusual, aspirational, or lifestyle appeal
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